

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250287617

Kind Code

A1

Publication Date

September 11, 2025

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SEMICONDUCTOR DEVICES AND METHODS FOR INCREASED CAPACITANCE

Abstract

Semiconductor devices having increased capacitance without increased fin height or increased chip area are disclosed. Grooves are formed across a width of the fin(s) to increase the overlapping surface area with the gate terminal, in particular with a length of the groove being less than or equal to the fin width. Methods of forming such grooved fins and semiconductor capacitor devices are also described.

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Family ID: 1000008620991

Appl. No.: 19/217058

Filed: May 23, 2025

Related U.S. Application Data

parent US continuation 17699300 20220321 parent-grant-document US 12336199 child US 19217058

Publication Classification

Int. Cl.: H10D1/66 (20250101); H01L21/265 (20060101); H10D1/00 (20250101)

U.S. Cl.:

CPC H10D1/66 (20250101); H01L21/26513 (20130101); H10D1/047 (20250101);

Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE [0001] This application is a divisional of U.S. patent application Ser. No. 17/699,300, filed on Mar. 21, 2022, now U.S. Patent No. //insert later//, which is incorporated by reference in its entirety.

BACKGROUND

[0002] Integrated circuits are formed on a semiconductor wafer. Photolithographic patterning processes use ultraviolet light to transfer a desired mask pattern to a photoresist on a semiconductor wafer. Etching processes may then be used to transfer the pattern to a layer below the photoresist. This process is repeated multiple times with different patterns to build different layers on the wafer substrate and make a semiconductor device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1A is a plan view of a fin field effect transistor (FinFET), in accordance with some embodiments.

[0005] FIG. 1B is a lateral side cross-sectional view along line B-B of FIG. 1A.

[0006] FIG. 2A is a plan view of a set of fins, showing an initial state, in accordance with some embodiments.

[0007] FIG. 2B is a longitudinal side cross-sectional view along line B-B of FIG. 2A.

[0008] FIG. 2C is a lateral side cross-sectional view along line C-C of FIG. 2A.

[0009] FIG. 3A is a plan view of the set of fins, showing a mask applied to form grooves in the fins.

[0010] FIG. 3B is a longitudinal side cross-sectional view along line B-B of FIG. 3A.

[0011] FIG. 3C is a lateral side cross-sectional view along line C-C of FIG. 3A.

[0012] FIG. 4A is a plan view of the set of fins, showing the fins with grooves formed across a width thereof.

[0013] FIG. 4B is a longitudinal side cross-sectional view along line B-B of FIG. 4A, illustrating a groove with a trapezoidal cross-section.

[0014] FIG. 4C is a lateral side cross-sectional view along line C-C of FIG. 4A.

[0015] FIG. 4D is an alternative longitudinal side cross-sectional view showing a groove with a rectangular cross-section.

[0016] FIG. 4E is an alternative longitudinal side cross-sectional view showing a groove with a triangular cross-section.

[0017] FIG. 5A is a plan view of the set of grooved fins, showing optional ion implantation.

[0018] FIG. 5B is a longitudinal side cross-sectional view along line B-B of FIG. 5A.

[0019] FIG. 5C is a lateral side cross-sectional view along line C-C of FIG. 5A.

[0020] FIG. 6A is a plan view of the set of grooved fins, showing application of an STI layer.

[0021] FIG. 6B is a longitudinal side cross-sectional view along line B-B of FIG. 6A.

[0022] FIG. 6C is a lateral side cross-sectional view along line C-C of FIG. 6A.

[0023] FIG. 7A is a plan view of the set of grooved fins, after etching of the STI layer.

[0024] FIG. 7B is a longitudinal side cross-sectional view along line B-B of FIG. 7A.

[0025] FIG. 7C is a lateral side cross-sectional view along line C-C of FIG. 7A.

[0026] FIG. 8A is a plan view of the set of grooved fins, showing the formation of a gate stack across the grooves in the fins.

[0027] FIG. 8B is a longitudinal side cross-sectional view along line B-B of FIG. 8A.

[0028] FIG. 8C is a lateral side cross-sectional view along line C-C of FIG. 8A.

[0029] FIG. 9A is a plan view of the set of grooved fins, showing the formation of gate spacers and a dielectric layer around the gate stack.

[0030] FIG. 9B is a longitudinal side cross-sectional view along line B-B of FIG. 9A.

[0031] FIG. 9C is a lateral side cross-sectional view along line C-C of FIG. 9A.

[0032] FIG. 10A is a plan view of the set of grooved fins, where the gate stack is a dummy gate stack and is subsequently removed, in accordance with some embodiments.

[0033] FIG. 10B is a longitudinal side cross-sectional view along line B-B of FIG. 10A.

[0034] FIG. 10C is a lateral side cross-sectional view along line C-C of FIG. 10A.

[0035] FIG. 11A is a plan view of the set of grooved fins, showing the formation of a new gate stack including a gate oxide layer and a gate layer, in accordance with some embodiments.

[0036] FIG. 11B is a longitudinal side cross-sectional view along line B-B of FIG. 11A.

[0037] FIG. 11C is a lateral side cross-sectional view along line C-C of FIG. 11A.

[0038] FIG. 12A is a plan view of the completed transistor, including vias and contacts for the source terminal, drain terminal, and the gate terminal, in accordance with some embodiments.

[0039] FIG. 12B is a longitudinal side cross-sectional view along line B-B of FIG. 12A.

[0040] FIG. 12C is a lateral side cross-sectional view along line C-C of FIG. 12A.

[0041] FIG. 13 is a flowchart showing the steps in a first method for forming a semiconductor device, in accordance with some embodiments herein.

[0042] FIG. 14A is a plan view showing a substrate with a mask applied thereon to form a set of grooved fins.

[0043] FIG. 14B is a longitudinal side cross-sectional view along line B-B of FIG. 14A.

[0044] FIG. 14C is a lateral side cross-sectional view along line C-C of FIG. 14A.

[0045] FIG. 15A is a plan view showing the substrate after etching to form the grooves and the fins concurrently.

[0046] FIG. 15B is a longitudinal side cross-sectional view along line B-B of FIG. 15A.

[0047] FIG. 15C is a lateral side cross-sectional view along line C-C of FIG. 15A.

[0048] FIG. 16A is a plan view showing the grooved fins after the mask has been removed.

[0049] FIG. 16B is a longitudinal side cross-sectional view along line B-B of FIG. 16A.

[0050] FIG. 16C is a lateral side cross-sectional view along line C-C of FIG. 16A.

[0051] FIG. 17A is a plan view of the set of grooved fins, showing application of an STI layer.

[0052] FIG. 17B is a longitudinal side cross-sectional view along line B-B of FIG. 17A.

[0053] FIG. 17C is a lateral side cross-sectional view along line C-C of FIG. 17A.

[0054] FIG. 18A is a plan view of the set of grooved fins, after etching of the STI layer.

[0055] FIG. 18B is a longitudinal side cross-sectional view along line B-B of FIG. 18A.

[0056] FIG. 18C is a lateral side cross-sectional view along line C-C of FIG. 18A.

[0057] FIG. 19A is a plan view of the set of grooved fins, with a mask applied to the ends of the fins and the portions of the STI layer within the exposed grooves further etched to reduce their depth, in accordance with some embodiments.

[0058] FIG. 19B is a longitudinal side cross-sectional view along line B-B of FIG. 19A.

[0059] FIG. 19C is a lateral side cross-sectional view along line C-C of FIG. 19A.

[0060] FIG. 20A is a plan view of the completed transistor, including vias and contacts for the source terminal, drain terminal, and the gate terminal, in accordance with some embodiments. The depth of the portions of the STI layer within the exposed grooves can also be used to control the capacitance by changing the overlapping surface area between the gate and the fins.

[0061] FIG. 20B is a longitudinal side cross-sectional view along line B-B of FIG. 20A.

[0062] FIG. 20C is a lateral side cross-sectional view along line C-C of FIG. 20A.

[0063] FIG. 21 is a flowchart showing the steps in a second method for forming a semiconductor device, in accordance with some embodiments herein.

[0064] FIG. 22A is a plan view of a set of grooved fins, divided into a capacitor region (left-hand side) and a normal region (right-hand side).

[0065] FIG. 22B is a longitudinal side cross-sectional view of the capacitor region along line B-B of FIG. 22A.

[0066] FIG. 22C is a lateral side cross-sectional view of the capacitor region along line C-C of FIG. 22A.

[0067] FIG. 22D is a longitudinal side cross-sectional view of the normal region along line D-D of FIG. 22A.

[0068] FIG. 22E is a lateral side cross-sectional view of the normal region along line E-E of FIG. 22A.

[0069] FIG. 23A is a plan view of the set of grooved fins, showing application of an STI layer.

[0070] FIG. 23B is a longitudinal side cross-sectional view of the capacitor region along line B-B of FIG. 23A.

[0071] FIG. 23C is a lateral side cross-sectional view of the capacitor region along line C-C of FIG. 23A.

[0072] FIG. 23D is a longitudinal side cross-sectional view of the normal region along line D-D of FIG. 23A.

[0073] FIG. 23E is a lateral side cross-sectional view of the normal region along line E-E of FIG. 23A.

[0074] FIG. 24A is a plan view of the set of grooved fins, showing application of a mask to the normal region and a subsequent first etching that affects only the capacitor region.

[0075] FIG. 24B is a longitudinal side cross-sectional view of the capacitor region along line B-B of FIG. 24A.

[0076] FIG. 24C is a lateral side cross-sectional view of the capacitor region along line C-C of FIG. 24A.

[0077] FIG. 24D is a longitudinal side cross-sectional view of the normal region along line D-D of FIG. 24A.

[0078] FIG. 24E is a lateral side cross-sectional view of the normal region along line E-E of FIG. 24A.

[0079] FIG. 25A is a plan view of the set of grooved fins, after the mask has been removed from the normal region.

[0080] FIG. 25B is a longitudinal side cross-sectional view of the capacitor region along line B-B of FIG. 25A.

[0081] FIG. 25C is a lateral side cross-sectional view of the capacitor region along line C-C of FIG. 25A.

[0082] FIG. 25D is a longitudinal side cross-sectional view of the normal region along line D-D of FIG. 25A.

[0083] FIG. 25E is a lateral side cross-sectional view of the normal region along line E-E of FIG. 25A.

[0084] FIG. 26A is a plan view of the set of grooved fins, after a subsequent second etching that reduces the depth of the STI layer in both the capacitor region and the normal region, exposing different amounts of the fins in each region.

[0085] FIG. 26B is a longitudinal side cross-sectional view of the capacitor region along line B-B of FIG. 26A.

[0086] FIG. 26C is a lateral side cross-sectional view of the capacitor region along line C-C of FIG. 26A.

[0087] FIG. 26D is a longitudinal side cross-sectional view of the normal region along line D-D of FIG. 26A.

[0088] FIG. 26E is a lateral side cross-sectional view of the normal region along line E-E of FIG. 26A.

[0089] FIG. 27A is a plan view of a third embodiment of a completed transistor, after formation of separate gate stacks in the two regions, gate spacers, and a dielectric layer, in accordance with some embodiments.

[0090] FIG. 27B is a longitudinal side cross-sectional view of the capacitor region along line B-B of FIG. 27A.

[0091] FIG. 27C is a lateral side cross-sectional view of the capacitor region along line C-C of FIG. 27A.

[0092] FIG. 27D is a longitudinal side cross-sectional view of the normal region along line D-D of FIG. 27A.

[0093] FIG. 27E is a lateral side cross-sectional view of the normal region along line E-E of FIG. 27A.

[0094] FIG. 28A is a plan view of a fourth embodiment of a completed transistor, where grooves are only present in the capacitor region and not the normal region, in accordance with some embodiments.

[0095] FIG. 28B is a longitudinal side cross-sectional view of the capacitor region along line B-B of FIG. 28A.

[0096] FIG. 28C is a lateral side cross-sectional view of the capacitor region along line C-C of FIG. 28A.

[0097] FIG. 28D is a longitudinal side cross-sectional view of the normal region along line D-D of FIG. 28A.

[0098] FIG. 28E is a lateral side cross-sectional view of the normal region along line E-E of FIG. 28A.

[0099] FIG. 29 is a flowchart showing the steps in a method for forming the third and fourth embodiments of a semiconductor device, in accordance with some embodiments herein.

DETAILED DESCRIPTION

[0100] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0101] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0102] The figures may provide views of fins across three different axes. The “plan view” is designated as being along the z-axis such that the length and width of the fin are visible, with the height extending into the page.

[0103] The “longitudinal side cross-sectional view” is arbitrarily designated as being along the x-axis such that the length and depth/height of the fin are visible, with the width extending into the

page.

[0104] The “lateral side cross-sectional view” is arbitrarily designated as being along the y-axis such that the width and depth/height of the fin are visible, with the length extending into the page.

[0105] Numerical values in the specification and claims of this application should be understood to include numerical values which are the same when reduced to the same number of significant figures and numerical values which differ from the stated value by less than the experimental error of conventional measurement technique of the type described in the present application to determine the value. All ranges disclosed herein are inclusive of the recited endpoint.

[0106] The term “about” can be used to include any numerical value that can vary without changing the basic function of that value. When used with a range, “about” also discloses the range defined by the absolute values of the two endpoints, e.g. “about 2 to about 4” also discloses the range “from 2 to 4.” The term “about” may refer to plus or minus 10% of the indicated number.

[0107] The present disclosure relates to semiconductor devices and methods for improving the performance of such devices. One type of three-dimensional design for a semiconductor device is a fin field effect transistor (FinFET), which is illustrated in plan view in FIG. 1A and in cross-sectional view in FIG. 1B. Referring first to FIG. 1A, the channel **100** between the source terminal **102** and the drain terminal **104** is in the form of thin vertical fins **120**. The source terminal **102** and the drain terminal **104** are illustrated as being in the same layer as the fins **120**, but can be located elsewhere (for example above the fins), depending on the three-dimensional design of the transistor/semiconductor device. It is common for a single FinFET to use several fins in parallel, covered by the same gate terminal. As better seen in FIG. 1B, the fin(s) **120** are formed from the same material as the substrate **110**. A dielectric layer **112** is then applied to the surfaces of the fins, and a conductive gate terminal **106** is thus present on three sides of each fin **120**.

[0108] The structure of the gate terminal **106** being separated from the fin/channel **120** by the dielectric layer **112** creates a capacitor. The capacitance is proportional to the overlapping surface area between the gate terminal and the fins. Thus, the capacitance can be increased by increasing the surface area of the gate terminal, or by increasing the height of the fins. However, when increasing the surface area of the gate terminal, the total area of the overall semiconductor device/chip may be increased as well, which is not desirable. Simply increasing the height of the fins can decrease their stability, causing undesirable cracking.

[0109] The present disclosure thus relates to semiconductor capacitor devices with increased capacitance without increasing the total area of the overall semiconductor device or increasing the height of the fins, and to methods for making such devices. The increased capacitance is obtained by forming one or more grooves across the width of the fins, or in other words by cutting into the fins. This increases the total overlapping surface area between the fins and the gate terminal, without increasing the height of the fin. This structure thus has a high capacitance density. This also permits the capacitance to be independently controlled without changing the fin height.

[0110] FIGS. 2A-12C illustrate a first method for forming such a semiconductor device, in accordance with some embodiments. It is noted that for clarity, the plan views do not include the areas between fins, although those areas are included in the longitudinal side cross-sectional views and the lateral side cross-sectional views. FIG. 13 is a flowchart summarizing the various steps in the first method, and will be further discussed after FIGS. 2A-12C.

[0111] Referring first to FIGS. 2A-2C, the fins **120** are illustrated here in an initial state. Each fin has an initial length **L0**, an initial width **W0**, and an initial depth or height **D0**. Although a rectangular shape is ideal for the fin, as can be seen in FIG. 2B and FIG. 2C, the fin may also have a generally trapezoidal cross-section, with potentially rounded sides and/or edges. The initial length **L0** and the initial width **W0** are measured at the top of the fin. The initial depth **D0** is also measured from the top of the fin.

[0112] In general embodiments, the initial fin length **L0** may range from about 20 nm to about 1200 nm. In general embodiments, the initial fin width **W0** may range from about 4 nm to about 20

nm. In general embodiments, the initial fin depth **D0** may range from about 10 nm to about 80 nm. Combinations of these ranges are contemplated.

[0113] The wafer substrate and the fins can be made of any semiconducting material. Such materials can include silicon, for example in the form of crystalline Si or polycrystalline Si. In alternative embodiments, the substrate and fins can be made of other elementary semiconductors such as germanium, or may include a compound semiconductor such as silicon carbide (SiC), gallium arsenide (GaAs), indium arsenide (InAs), and indium phosphide (InP).

[0114] The fins **120** may be formed by any appropriate process. For example, self-aligned double patterning (SADP) or self-aligned quadruple patterning (SAQP) can be used to form the fins. Briefly, a hard mask is deposited upon the substrate and then patterned to form a mandrel. Next, a spacer is formed by deposition upon the horizontal and vertical surfaces of the mandrel. The horizontal surfaces of the spacer are then etched, leaving only the spacer on the two vertical surfaces of the mandrel. The mandrel is then removed, leaving only the two vertical spacers. This doubles the line density upon the substrate (i.e. SADP) compared to the original pattern, or halves the line pitch. These SADP steps can be repeated using the two vertical spacers as a new mandrel to halve the pitch again (i.e. SAQP). The spacers are then used as a mask to perform etching of the substrate and obtain the fins.

[0115] Referring next to FIGS. **3A-3C**, a patterned mask **130** is applied to the fins **120**. The mask is typically formed using photoresist, although a hard mask material, such as a nitride, could also be used. As illustrated here by way of example, the fins are exposed in three areas **132**, which will correspond to grooves that are formed across the width of each fin. The fins are then etched to form the grooves. This may be done, for example, using wet etching processes, dry etching processes, or plasma etching.

[0116] Referring now to FIGS. **4A-4C**, the patterned mask has been removed, and the etched fin **120** with fin cuts or grooves is shown. In the plan view of FIG. **4A**, dotted rectangles are used to indicate the location of the grooves **140** on the fins **120**.

[0117] As illustrated here, three grooves **140** are formed which extend across the width of the fin structure. The length, width, and depth of each groove is measured in the same direction as the length, width, and depth of the fin. The three grooves have the same depth, which is indicated as depth D_g . The initial fin depth **D0** is also shown. In particular embodiments, the groove depth D_g can be deeper or shallower than the initial fin depth **D0**. Put another way, D_g can be greater than or less than **D0**. In particular embodiments, and as illustrated here, D_g is less than **D0**.

[0118] The length and depth of the grooves can differ between each other, which is determined by the patterned mask. For illustrative purposes, the length of one groove at the top of the fin is indicated as length L_g . When L_g is less than or equal to **W0**, there are multiple combinations of groove length and groove depth for which the surface area within the new groove is greater than the surface area of the portion of the original fin which was removed to form the groove. As a result, then, the capacitance can be increased to a desired value by varying the total number of grooves and their groove length and groove depth.

[0119] As illustrated in FIG. **4B**, each groove **140** also has a trapezoidal shape, with two sides **142**, **144** and a floor **146**. The shape of the groove may vary depending on the etching method, and for example may be rectangular, or triangular (i.e. without a floor), or trapezoidal. An illustrative example of a fin **120** with rectangular grooves **140** is provided in FIG. **4D**, and an illustrative example of a fin **120** with triangular grooves **140** is provided in FIG. **4E**.

[0120] Referring now to FIGS. **5A-5C**, in an optional step, ion implantation is performed on the fin **120** (indicated with arrows). This may be done, for example, to set the threshold voltage to a desired value, or to modify other behavior or properties. Common p-type dopants may include boron or gallium. Common n-type dopants may include phosphorus or arsenic. The ion implantation is also illustrated in FIG. **5C** as being performed in the areas between fins, but this is not required, and ion implantation can be performed solely on the fins using appropriate masking of

the areas between fins.

[0121] Continuing in FIGS. **6A-6C**, a shallow trench isolation (STI) layer **150** is deposited over the fins **120**. The STI layer is formed from a dielectric material, commonly silicon dioxide, although other dielectric materials can also be used such as undoped polysilicon, silicon oxide, silicon nitride, silicon oxynitride, fluoride-doped silicate glass, or other low-k dielectric material. The deposition can be done using physical vapor deposition (PVD) or chemical vapor deposition (CVD) or spin-on processes known in the art, or can be grown via oxidation. As seen in FIG. **5B**, the STI layer also fills the grooves **140**.

[0122] Continuing with FIGS. **7A-7C**, the STI layer **150** is etched. This can be done, for example, using wet etching processes, dry etching processes, or plasma etching. As illustrated here, the STI layer is etched to a level below the grooves **140**, such that the STI layer is not present within the grooves, and fills only the areas between fins to form isolation regions **152** between adjacent fins. As will be seen further herein, in other embodiments, the STI layer can be present within the grooves.

[0123] Next, in FIGS. **8A-8C**, a first gate stack **160** is formed. As seen in FIG. **8B**, the gate stack **160** fills the grooves **140**. As seen in FIG. **8C**, the gate stack **160** also fills the areas between adjacent fins **120**. As illustrated here, the gate stack **160** includes a gate oxide layer **162** adjacent the fin **120**, and a gate layer **164** upon the gate oxide layer **162**. An optional cap layer **166** upon the gate layer **164** is also illustrated.

[0124] The gate oxide layer may be any dielectric material, for example silicon dioxide, hafnium silicate, zirconium silicate, hafnium dioxide, or zirconium dioxide. The gate oxide layer **162** may be very thin compared to the gate layer **164**. The gate oxide layer may be formed by a process that produces a thin film, such as atomic layer deposition (ALD). Alternatively, the gate oxide layer can be grown by thermal oxidation of the substrate, for example at a temperature of about 850° C. to about 950° C. in the presence of water or oxygen (O.sub.2). As yet another example, the gate oxide layer could be formed by a chemical vapor deposition (CVD) process, for example using O.sub.2 along with silane (SiH.sub.4) or dichlorosilane (SiH.sub.2Cl.sub.2), or using tetraethyl orthosilicate (TEOS) at elevated temperatures above about 600° C. As yet another example, the gate oxide layer could be formed by the decomposition of TEOS at temperatures of about 600° C. to about 650° C., or through plasma enhanced CVD at lower temperatures.

[0125] The gate layer may be any electrically conductive material, for example aluminum, polysilicon (doped or undoped), tungsten, a metal silicide such as TiSi or MoSi.sub.2 or TaSi or WSi.sub.2, or material such as Ta, TaN, Nb, WN, or WN/RuO.sub.2. The cap layer **166**, when present, is used as a mask and can be made of appropriate materials such as silicon oxide, silicon nitride, silicon oxynitride, or other metal. The gate layer **164** and the optional cap layer **166** may be formed by PVD or CVD, or by other processes such as sputtering.

[0126] As will be seen further below, this gate stack may be a dummy gate stack which is subsequently removed and replaced, or may be used as the gate terminal in the final semiconductor device. The three layers may each be individually deposited upon the substrate, then a patterned mask is applied (e.g. photoresist) and the layers are etched to obtain the gate stack in the desired location. Wet etching processes, dry etching processes, or plasma etching may be used as desired for etching of each layer.

[0127] In FIGS. **9A-9C**, gate spacers **170** and a dielectric layer **180** are formed upon the exposed portions of the fins **120** and the STI layer **150**. The fins **120** are shown in dashed line outline. The gate spacers act as an electrical insulator, and can be made from any oxide, such as silicon dioxide (which may be doped with fluorine or carbon). The gate spacer can be formed by a deposition process (e.g. PVD or CVD) on the entire surface, then applying a patterned mask and etching to remove material from undesired locations. The dielectric layer **180** may be any suitable material, such as silicon dioxide (which may be doped with fluorine or carbon).

[0128] In some embodiments, the gate stack **160** acts as a dummy stack that provides support for

forming the gate spacers **170**. In FIGS. **10A-10C**, the gate stack **160** is removed. For example, a patterned mask can be placed over the gate spacers **170** and the dielectric layer **180**, leaving the gate stack exposed, and the gate stack is then removed using an appropriate etching process.

[0129] In FIGS. **11A-11C**, a new gate stack **190** is then formed. As illustrated here, the new gate stack includes a gate oxide layer **192** and a gate layer **194**. Compared to the gate oxide layer **162** of FIGS. **8A-8C**, the gate oxide layer also runs up the side of the gate spacers **170**, and can be formed using ALD. Again, the gate oxide layer may be any dielectric material, for example silicon dioxide, hafnium silicate, zirconium silicate, hafnium dioxide, or zirconium dioxide. The gate layer **194** can be, for example, a single-layer structure formed from a single metal, or can be a multi-layer structure containing different metals. The gate layer can be deposited using methods such as PVD or CVD or sputtering. Again, the steps shown in FIGS. **10A-11C** are optional, and the original gate stack formed in FIGS. **8A-8C** can be maintained in the final device if desired.

[0130] Finally, in FIGS. **12A-12C**, additional layers are formed to complete the semiconductor device/transistor **200**. An insulation layer **210** is deposited over the gate stack **190** and the dielectric layer **180**. As best seen in FIG. **12B**, vias **212** are etched through the insulation layer **210** and the dielectric layer **180** down to the fins **120**, and then filled with an electrically conductive material which is also used to form contacts **214** for the source terminal and the drain terminal. As best seen in FIG. **12C**, a via **222** is also etched through the insulation layer **210** down to the gate stack **190**, then filled with an electrically conductive material which is also used to form a contact **224** for the gate terminal. The insulation layer is electrically insulating, and can be made from any appropriate material.

[0131] FIG. **13** is a flowchart summarizing the various steps illustrated in FIGS. **2A-12C**.

[0132] In step **300**, a substrate with at least one fin is received. The substrate usually has a plurality of fins thereon. This is illustrated in FIGS. **2A-2C**.

[0133] In step **305**, a mask with a pattern is applied to the substrate. The pattern extends across the width of the fin(s), and exposes one or more portions of the fin. This is illustrated in FIGS. **3A-3C**.

[0134] In step **310**, the fin/substrate is etched. This forms a groove in the fin which extends across the entire width of the fin. Multiple grooves can be formed in this step if desired. In step **315**, the mask is removed. The result after this step is illustrated in FIGS. **4A-4C**.

[0135] In optional step **317**, ion implantation is performed on the fin(s). This is illustrated in FIGS. **5A-5C**.

[0136] In step **320**, an STI layer is applied to the substrate. The STI layer may cover the fins. This is illustrated in FIGS. **6A-6C**. In step **325**, the STI layer is etched to form isolation regions between fins. This is illustrated in FIGS. **7A-7C**.

[0137] In step **330**, a first gate stack is formed which fills the groove(s) within the fin(s). The layers within the gate stack include a gate oxide layer, a gate layer, and an optional cap layer. This is illustrated in FIGS. **8A-8C**.

[0138] In step **335**, gate spacers are formed adjacent the first gate stack. In step **340**, a dielectric layer is formed adjacent the gate spacers, such that the gate spacers separate the dielectric layer from the first gate stack. This is illustrated in FIGS. **9A-9C**.

[0139] In optional step **341**, the first gate stack is removed. This is illustrated in FIGS. **10A-10C**. In optional step **343**, a second or new gate stack is formed. The second gate stack may comprise a gate oxide layer and a metal gate layer. This is illustrated in FIGS. **11A-11C**. It is noted that these steps are optional, and the first gate stack may be maintained within the final semiconductor device if desired.

[0140] In step **345**, an insulation layer is formed over the gate stack and the dielectric layer. In step **350**, vias are etched through the insulation layer and the dielectric layer, exposing portions of the fin(s). In step **355**, the vias are filled with an electrically conductive material. In step **360**, a via is etched through the insulation layer, exposing a portion of the gate stack. In step **365**, the via is filled with an electrically conductive material. These vias form contacts for the source terminal, the

drain terminal, and the gate terminal. It is noted that steps **350**, **355**, **360**, **365** can be performed in any desired order. The result is illustrated in FIGS. **12A-12C**.

[0141] FIGS. **14A-16C** illustrate a second method for forming such a semiconductor device, in accordance with some embodiments. In this second embodiment, the grooves within the fins are formed at the same time as the fins. Put another way, the same photolithographic and etching steps are also used to form the grooves.

[0142] Initially, FIGS. **14A-14C** illustrate the substrate **110** with a patterned mask **400** thereon. The substrate **110** is exposed in all areas except where the fins are desired to be placed or located. The entire substrate, including the areas between where the fins will be etched, are shown in all three views. The patterned mask **400** includes one or more spacer lines **402**; multiple spacer lines are shown here. Each spacer line **402** includes at least one gap **404** (indicated as a dotted box); three gaps are illustrated in each spacer line in this example. The length of each gap L_g is less than or equal to a width of the spacer line S_w . The spacer lines correspond to the areas on the substrate that will become fins. The spacer line gaps **404** correspond to the groove(s) that will be formed in the fins.

[0143] FIGS. **15A-15C** illustrate the substrate **110** after etching, with the patterned mask **400** still applied thereon. The plan view of FIG. **15A** has dotted rectangles indicating where the grooves **140** have been formed. The grooves are more visible in FIG. **15B**. The fins **120** are now visible in FIG. **15C**, along with the areas **406** between fins.

[0144] In FIGS. **16A-16C**, the patterned mask has been removed. In the plan view of FIG. **16A**, the substrate is no longer shown between the fins **120** for clarity. Dotted rectangles are used again to indicate the location of the grooves **140** on the fins **120**. In this second embodiment, the groove depth D_g is substantially equal to the initial fin depth D_0 . The resulting structure is substantially similar to the structure depicted in FIGS. **4A-4C**.

[0145] Continuing then, FIGS. **17A-20C** illustrate some alternative processing steps which can also be used to prepare a semiconductor device.

[0146] In FIGS. **17A-17C**, a shallow trench isolation (STI) layer **150** is deposited over the fins **120**. As seen in FIG. **17B**, the STI layer also fills the grooves **140**. The resulting structure is substantially similar to the structure depicted in FIGS. **6A-6C**.

[0147] In FIGS. **18A-18C**, the STI layer is then etched. However, in contrast to the structure of FIGS. **7A-7C**, the STI layer is etched such that the STI layer is still present within the grooves (indicated with reference numeral **154**).

[0148] Next, as illustrated in FIGS. **19A-19C**, a patterned mask **410** (e.g. photoresist) can be applied to the ends of the fin(s) **120**, leaving the grooves **140** exposed. Further etching is then performed, such that the depth of the STI portions **154** within the grooves S_g and in the area where the gate stack will be formed is less than the depth of the STI portions in the area where the dielectric layer will be formed S_d .

[0149] FIGS. **20A-20C** illustrate the resulting transistor **200** after the gate oxide layer **192**, gate layer **194**, gate spacers **170**, dielectric layer **180**, insulation layer **210**, vias **212**, **222**, and contacts **214**, **216**, **224** are applied as described in FIGS. **8A-12C** and in the flowchart of FIG. **13**. The presence of the STI layer between the fins and the gate oxide layer provides another way to control the capacitance, by changing the overlapping surface area between the gate stack and the fins.

[0150] FIG. **21** is a flowchart summarizing the various steps illustrated in FIGS. **14A-20C**.

[0151] In step **505**, a mask with a pattern is applied to the substrate. The pattern includes one or more spacer lines, corresponding to the number of fins to be formed. Each spacer line has one or more gaps which will correspond to the grooves. The length of each gap is less than or equal to the width of the spacer line. This is illustrated in FIGS. **14A-14C**.

[0152] In step **510**, the substrate is etched. This forms one or more fins, which have one or more grooves that extend across the width of the fin. This is illustrated in FIGS. **15A-15C**.

[0153] In step **515**, the mask is removed. The result after this step is illustrated in FIGS. **16A-16C**.

[0154] In optional step **517**, ion implantation is performed on the fin(s). This step corresponds to that previously illustrated in FIGS. **5A-5C**, and is not illustrated again.

[0155] In step **520**, an STI layer is applied to the substrate. The STI layer may cover the fins. This is illustrated in FIGS. **17A-17C**. In step **525**, the STI layer is etched to form isolation regions between fins. Portions of the STI layer are still present within the grooves. This is illustrated in FIGS. **18A-18C**.

[0156] Next, in step **530**, a mask is applied to the ends of the fin(s), leaving the grooves exposed. In step **535**, the grooves are further etched so that the depth of the STI portions within the grooves is reduced compared to the depth of the STI portions in the area beyond the ends of the fins. The result after this step is illustrated in FIGS. **19A-19C**.

[0157] Next, in step **540**, a first gate stack is formed which fills the groove(s) within the fin(s). The layers within the gate stack include a gate oxide layer, a gate layer, and an optional cap layer. In step **545**, gate spacers are formed adjacent the first gate stack. In step **550**, a dielectric layer is formed adjacent the gate spacers, such that the gate spacers separate the dielectric layer from the first gate stack. These steps correspond to those previously illustrated in FIGS. **8A-9C**, and are not illustrated again.

[0158] In optional step **551**, the first gate stack is removed. In optional step **553**, a second or new gate stack is formed. The second gate stack may comprise a gate oxide layer and a metal gate layer. Again, these steps are optional, and the first gate stack may be maintained within the final semiconductor device if desired. These steps correspond to those previously illustrated in FIGS. **10A-11C**, and are not illustrated again.

[0159] In step **555**, an insulation layer is formed over the gate stack and the dielectric layer. In step **560**, vias are etched through the insulation layer and the dielectric layer, exposing portions of the fin(s). In step **565**, the vias are filled with an electrically conductive material. In step **570**, a via is etched through the insulation layer, exposing a portion of the gate stack. In step **575**, the via is filled with an electrically conductive material. These vias form contacts for the source terminal, the drain terminal, and the gate terminal. The steps for forming the contacts **560**, **565**, **570**, **575** can be performed in any order. The result is illustrated in FIGS. **20A-20C**.

[0160] FIGS. **22A-28E** illustrate a third method and a fourth method for forming a semiconductor device, in accordance with some embodiments. In these two methods, the semiconductor device includes two separate transistors, which have different capacitances. The transistor with a higher capacitance is denoted as a “capacitor region”, and the transistor with a lower capacitance is denoted as a “normal region”. Again, for clarity, the plan views do not include the areas between fins. FIG. **29** is a flowchart summarizing the various steps in these two methods, and will be further discussed after FIGS. **21A-28E**.

[0161] Beginning with FIGS. **22A-22E**, the fins **120** are shown. The fins extend across both the capacitor region **600** and the normal region **610**, as seen in FIG. **22A**. It is contemplated that the fin length, fin width, and fin height/depth are the same in both regions.

[0162] Grooves **122** are present in the capacitor region, and grooves **124** are also present in the normal region. The grooves in the capacitor region have depth D_c and length L_c . The grooves in the normal region have depth D_n and length L_n . As illustrated here, the two depths D_c , D_n are equal, and the two lengths L_c , L_n are also equal. The longitudinal side cross-sectional views and the lateral side cross-sectional views are the same for both regions, as seen in FIGS. **22B-22E**.

[0163] In other embodiments, it is contemplated that the depth D_c is greater than the depth D_n and/or the length L_c is greater than the length L_n . As a result, the surface area of the grooves **122** in the capacitor region is greater than the surface area of the grooves **124** in the normal region. Again, it is noted that the two lengths L_c , L_n are also less than or equal to the fin width W_0 . The groove depth D_c of the grooves in the capacitor region can be made different from the groove depth D_n of the grooves in the normal region, for example, by applying a mask (e.g. photoresist) to expose one region and not the other region, so that only the grooves in the exposed region are etched. This

permits the groove depth to be independently controlled.

[0164] Next, in FIGS. **23A-23E**, an STI layer **150** is deposited over the fins. The STI layer also fills the grooves **122**, **124** in both regions. The portion of the STI layer in the capacitor region is marked with reference numeral **652**, and the portion of the STI layer in the capacitor region is marked with reference numeral **654**.

[0165] Then, in FIGS. **24A-24E**, a mask **620** (for example, photoresist) is applied to only the normal region **610** and not the capacitor region **600**. The STI layer **150** is then etched. Due to the presence of the mask, only the STI layer in the capacitor region **652** is etched. As a result, the depth of the STI layer in the capacitor region **652** is less than the depth of the STI layer in the normal region **654**. This intermediate depth of the STI layer in the capacitor region is indicated as S_{ci} , while the depth of the STI layer in the normal region is indicated as S_n .

[0166] The mask **620** is then removed, resulting in the structure shown in FIGS. **25A-25E**. In FIGS. **26A-26E**, the STI layer **150** is etched again, reducing the thickness of the STI layer in both regions **652**, **654**, so that the fins are exposed in both regions. The final depths of the STI layer in each region are now indicated as depths S_{cf} , S_{nf} .

[0167] FIGS. **27A-27E** illustrate the two regions after the dielectric layer **180**, gate spacers **170**, gate oxide layer **192**, and gate layer **194** have been deposited and patterned to form a first gate stack **660** in the capacitor region and a second gate stack **670** in the normal region, thus forming two separate transistors, one in each region. As seen in the plan view of FIG. **27A**, the dielectric layer **180** electrically isolates the two transistors from each other. The fins **120** are shown as dashed lines. As seen by comparing FIGS. **27B-27C** with FIGS. **27D-27E**, the transistor in the capacitor region has a higher capacitance than the transistor in the normal region, due to the higher overlapping surface area between the fins **120** and the first gate stack **660**.

[0168] FIGS. **28A-28E** illustrate the fourth method for forming a semiconductor device. The device of FIGS. **28A-28E** differs from that of FIGS. **27A-27E** in that no grooves are present in the fins in the normal region **610**, as best seen in FIG. **28D**. Again, the result of this structure is that the transistor in the capacitor region has a higher capacitance than the transistor in the normal region. This device can be formed by omitting the creation of grooves in the portions of the fins located in the normal region.

[0169] The devices of FIGS. **27A-28E**, having transistors with different capacitances, may act as a CMOS circuit. This may be useful for chips used in image processing or other applications.

[0170] In additional embodiments, other structures can be changed between the capacitor region and the normal region to change their capacitances. As already described, the grooves between the two regions could have different depths, or have STI layers with different depths. As another example, it is contemplated that the gate oxide layer in the capacitor region and the normal region can be different from each other. For example, the two gate oxide layers could be made from different materials that have a different dielectric constant, or the two gate oxide layers could have different thicknesses. As yet another example, the fins in the two regions could have different doping concentrations, and/or use different dopants.

[0171] FIG. **29** is a flowchart summarizing the various steps illustrated in FIGS. **22A-28E**.

[0172] In step **705**, a substrate with at least one fin is received. The substrate usually has a plurality of fins thereon. At least one groove is present in the fin, which extends across the entire width of the fin. As illustrated in FIGS. **22A-22E**, a plurality of grooves is present, with at least a first groove in a capacitor region and a second groove in a normal region.

[0173] In optional step **707**, ion implantation is performed on the fin(s). It is noted that the ions implanted in the capacitor region and the normal region may be the same, or may be different. In more particular embodiments, it is contemplated that one region is doped with a p-type dopant and the other region is doped with an n-type dopant. This step corresponds to that previously illustrated in FIGS. **5A-5C**, and is not illustrated again.

[0174] In step **710**, an STI layer is applied to the substrate. The STI layer may cover the fins, and is

present in both the capacitor region and the normal region. This is illustrated in FIGS. 23A-23E. In step 715, a mask is applied to only the normal region. A mask is not applied to the capacitor region, so the STI layer in the capacitor region remains exposed. In step 720, the STI layer in the capacitor region is etched to an intermediate depth. This is illustrated in FIGS. 24A-24E.

[0175] In step 725, the mask is removed. This is illustrated in FIGS. 25A-25E. In step 730, the STI layer is etched again. This etching reduces the depth of the STI layer in both the capacitor region and the normal region, so that the fins are exposed. This is illustrated in FIGS. 26A-26E.

[0176] Next, in step 735, a first gate stack is formed in the capacitor region, which fills the fin groove(s) present in the capacitor region. A second gate stack is also formed in the normal region, which fills any fin groove(s) present in the normal region (compare FIGS. 27D-27E with FIGS. 28D-28E). In step 740, gate spacers are formed adjacent the two gate stacks. In step 745, a dielectric layer is formed adjacent the gate spacers, such that the gate spacers separate the dielectric layer from the two gate stacks. These steps correspond to those previously illustrated in FIGS. 8A-9C, and are not illustrated again.

[0177] In optional step 747, one or both gate stacks are removed. In optional step 749, one or two new gate stacks are formed. The new gate stacks may comprise a gate oxide layer and a metal gate layer. Again, these steps are optional, and the original gate stacks may be maintained within the final semiconductor device if desired. These steps correspond to those previously illustrated in FIGS. 10A-11C, and are not illustrated again.

[0178] In step 750, an insulation layer is formed over the two gate stacks and the dielectric layer. In step 755, vias are etched through the insulation layer and the dielectric layer in both the capacitor region and the normal region, exposing portions of the fin(s). In step 760, vias are etched through the insulation layer, exposing a portion of the gate stack in both the capacitor region and the normal region. In step 765, the vias are filled with an electrically conductive material to form contacts for the source terminal, the drain terminal, and the gate terminal. The steps for forming the contacts 755, 760, 765 can be performed in any order. These steps correspond to those previously illustrated in FIGS. 12A-12C (suitably modified), and are not illustrated again.

[0179] The semiconductor devices described herein, having fins with grooves across the fin width, provide higher capacitance compared to fins without such grooves. This provides the advantage of being able to independently control the capacitance without having to change the fin height or change the overall chip area. In addition, the grooves can be made without needing to add extra processing steps to the overall chip manufacturing process. Additional parameters, such as by changing the presence or depth of an STI layer within the grooves, or by varying properties of the gate oxide layer, can also be used to provide further control over the capacitance.

[0180] Some embodiments of the present disclosure thus relate to semiconductor devices. The devices comprise a substrate that comprises at least one fin. The fin has at least a first groove across a width of the fin. A first gate stack is disposed over the at least one fin, and fills the first groove. A first source and a first drain are each electrically connected to the at least one fin.

[0181] Also disclosed in various embodiments are methods for forming a semiconductor capacitor device. A fin is formed on a substrate. A mask with a pattern across a width of the fin is then applied. The fin is etched to form at least a first groove across the width of the fin. The mask is removed. A first gate stack is formed in the first groove of the fin.

[0182] Other embodiments disclosed herein include other methods for forming a semiconductor capacitor device. A first mask with a pattern is applied to a substrate. The first mask pattern comprises a spacer line that includes at least one gap, wherein the length of each gap is less than or equal to the width of the spacer line. The substrate is then etched to form a fin with at least one groove across the width of the fin. The first mask is then removed.

[0183] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other

processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor device, comprising: a substrate comprising at least one fin, wherein the fin comprises at least a first groove across an entire width of the fin and a second groove across the entire width of the at least one fin, wherein the first groove and the second groove have different depths; a first gate stack disposed over the at least one fin, wherein the first gate stack fills the first groove; and a first source and a first drain, each electrically connected to the at least one fin along a length of the fin; wherein the first gate stack, the first source, the first drain, and the at least one fin form a transistor
2. The semiconductor device of claim 1, wherein the first groove has a depth that is deeper or shallower than a height of the at least one fin.
3. The semiconductor device of claim 1, wherein in a longitudinal side cross-sectional view, the first groove has a triangular or trapezoidal or rectangular shape.
4. The semiconductor device of claim 1, wherein the first groove and the second groove each include an STI layer, wherein the depths of the STI layers differ from each other.
5. The semiconductor device of claim 1, wherein the at least one fin passes through a capacitor region and a normal region, and the first groove is in the capacitor region; and wherein a second gate stack is formed in the normal region which is electrically isolated from the first gate stack.
6. The semiconductor device of claim 5, wherein an STI layer in the capacitor region has a different depth than an STI layer in the normal region.
7. The semiconductor device of claim 5, wherein a gate oxide layer of the first gate stack is different from a gate oxide layer of the second gate stack.
8. The semiconductor device of claim 1, wherein the first gate stack comprises a gate oxide layer and a gate layer.
9. The semiconductor device of claim 1, further comprising an STI layer at a bottom of the first groove between the at least one fin and the first gate stack.
10. The semiconductor device of claim 1, further comprising gate spacers adjacent the first gate stack.
11. The semiconductor device of claim 1, further comprising an insulation layer above the first gate stack, and electrical contacts passing through the insulation layer to the first gate stack and the at least one fin.
12. A semiconductor device, comprising: a substrate comprising at least one fin, wherein the fin comprises at least a first groove across an entire width of the fin; a first gate stack disposed over the at least one fin, wherein the first gate stack fills the first groove; and a first source and a first drain, each electrically connected to the at least one fin along a length of the fin; wherein the first gate stack, the first source, the first drain, and the at least one fin form a transistor; wherein the at least one fin passes through a capacitor region and a normal region, and the first groove is in the capacitor region; and wherein a second gate stack is formed in the normal region which is electrically isolated from the first gate stack; and wherein an STI layer in the capacitor region has a different depth than an STI layer in the normal region.
13. The semiconductor device of claim 12, wherein the at least one fin further comprises a second groove across the width of the at least one fin.
14. The semiconductor device of claim 13, wherein the first groove and the second groove each include an STI layer, wherein the depths of the STI layers differ from each other.

15. The semiconductor device of claim 13, wherein the first groove and the second groove have different depths.

16. The semiconductor device of claim 12, wherein a gate oxide layer of the first gate stack is different from a gate oxide layer of the second gate stack.

17. A semiconductor device, comprising: a substrate comprising at least one fin, wherein the fin comprises at least a first groove across an entire width of the fin; an STI layer, wherein a top of the at least one fin extends above the STI layer; a first gate stack disposed over the at least one fin, wherein the first gate stack fills the first groove; and a first source and a first drain, each electrically connected to the at least one fin along a length of the fin; wherein the first gate stack, the first source, the first drain, and the at least one fin form a transistor.

18. The semiconductor device of claim 17, wherein the first groove has a depth that is deeper or shallower than a height of the at least one fin.

19. The semiconductor device of claim 17, wherein the at least one fin further comprises a second groove across the width of the at least one fin.

20. The semiconductor device of claim 17, wherein the at least one fin passes through a capacitor region and a normal region, and the first groove is in the capacitor region; and wherein a second gate stack is formed in the normal region which is electrically isolated from the first gate stack.
